

Combinatorics PhD Exam, August 2023

1. Prove that for all positive integers n , the identity

$$n \binom{2n-1}{n-1} = \sum_{k=1}^n k \binom{n}{k}^2$$

holds.

2. Let B_n be the number of compositions of n in which the first part is odd. Find a closed-form, explicit formula (no summation signs) for $B_n + B_{n+1}$.
3. How many ways are there to choose a permutation of length n and color each of its cycles red, blue, or green? Give a closed-form, explicit formula (no summation signs) in terms of n .
4. Let w_n be the number of words of length n over the alphabet $\{A, B, C\}$ in which a B is not immediately followed by a C . Find the explicit form of the ordinary generating function of the sequence $w_0, w_1, w_2, \dots$.
5. This problem has two parts.
- (a) For which nonnegative integers is the Catalan number C_n odd?
 - (b) Let $a_0, a_1, a_2, \dots$ be an infinite sequence that is polynomially recursive. Let b_i be the remainder of a_i modulo 2. Does it follow that the b_i form a polynomially recursive sequence?
6. Prove that an r -regular graph of girth (length of the shortest cycle) four has at least $2r$ vertices, with equality only for the complete bipartite graph $K_{r,r}$.
7. Let a and b be positive integers. We say that a is a unitary divisor of b if a divides b and a and b/a are relatively prime to each other. Let P be the poset of all positive integers in which $a \preceq b$ if a is a unitary divisor of b . Describe the Möbius function of P . That is, provide a formula for $\mu_P(a, b)$.
8. Prove that every graph with m edges contains a bipartite subgraph with at least $m/2$ edges. (One approach is to compute the expected number of edges in a random bipartite subgraph, but there are also non-probabilistic proofs.)